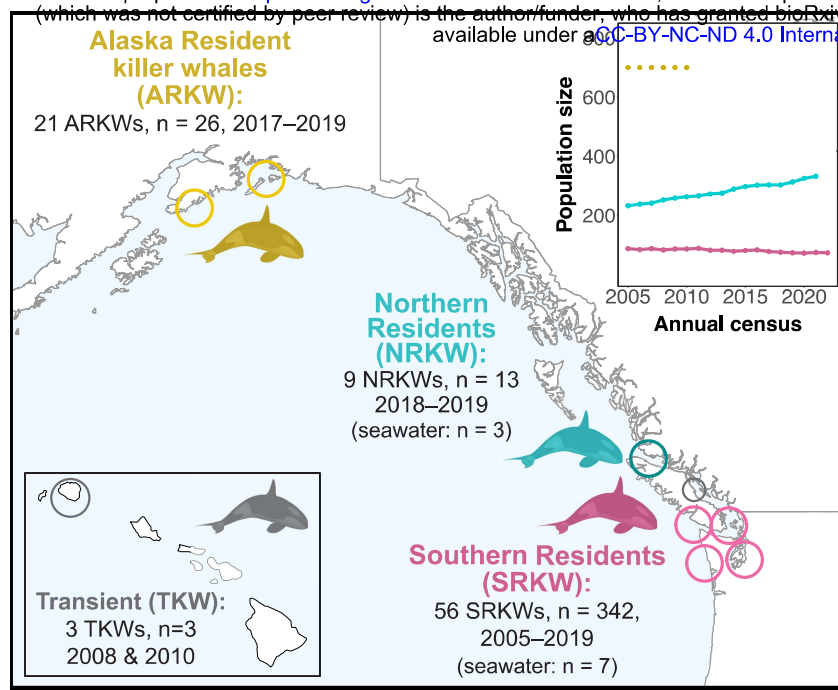


Stress survival	<i>dnaK</i>	Stress tolerance	90.30%	92.23%
Anti-phagocytosis	<i>galE</i>	Capsule biosynthesis	85.10%	87.12%
VF regulation	<i>cvfB</i>	Hemolysin production	84.80%	87.10%
T3SS	<i>ctpA</i>	Cytotoxicity	83.50%	
Biofilm	<i>dnaJ</i>	Biofilm, desiccation tolerance	82.60%	86.89%
Adherence	<i>yiaD (OmpA)</i>	Host cell-binding porin	82.60%	84.50%
Metabolism	<i>pfo</i>	Pyruvate metabolism	81.10%	93.70%
Immune evasion	<i>kpsF</i>	Capsule biosynthesis, anti-phagocytosis	80.90%	76.77%
Pore-forming toxins	<i>hlyC/corC</i>	Hemolysin; host cell lysis	80.30%	83.78%
Motility	<i>fliY</i>	Flagellum-mediated motility	79.70%	82.36%
Pore-forming toxins	<i>hemolysin</i>	Hemolysin; host cell lysis	78.90%	77.29%
Stress survival	<i>coa_dr</i>	ROS detoxification	77.88%	81.85%
Adherence	<i>ompA</i>	Host cell-binding porin	77.40%	
Adherence	<i>upaG (YadA)</i>	Fibronectin and laminin-binding adhesin	77.20%	
Anti-phagocytosis	<i>murJ</i>	Capsule/peptidoglycan	77.10%	72.92%
Pore-forming toxins	<i>tlyA</i>	Hemolysin; host cell lysis	75.70%	84.93%
Adherence	<i>ompH</i>	Host cell-binding periplasmic chaperone	75.00%	
Pore-forming toxins	<i>hlyD</i>	Hemolysin; host cell lysis	73.60%	73.42%
Immune evasion	<i>brkB</i>	Resists complement-mediated killing	73.00%	
T4SS	<i>virB4</i>	T4SS protein; replication within host	70.80%	
Invasion	<i>MORN repeat</i>	Outer membrane transport protein	70.40%	
T4SS	<i>virB11</i>	T4SS ATPase; replication within host	69.20%	
Pore-forming toxins	<i>shlB</i>	Hemolysin transporter	69.00%	
Pore-forming toxins	<i>cirA</i>	Cytotoxic activity	68.80%	78.03%
Exotoxins	<i>epsF</i>	T2SS toxin secretion	64.79%	84.54%
Stress survival	<i>creD</i>	Cell envelope integrity		82.68%
Pore-forming toxins	Septicolysin	Cytolysin		82.57%
VF regulation	<i>TidD</i>	Regulates hemolysin		80.19%

* Full results, references and sequences in **Supplementary Table 10 and Supplementary Data 9**

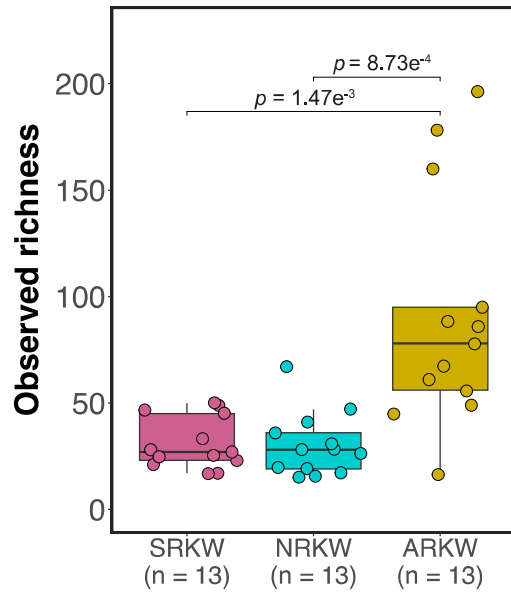
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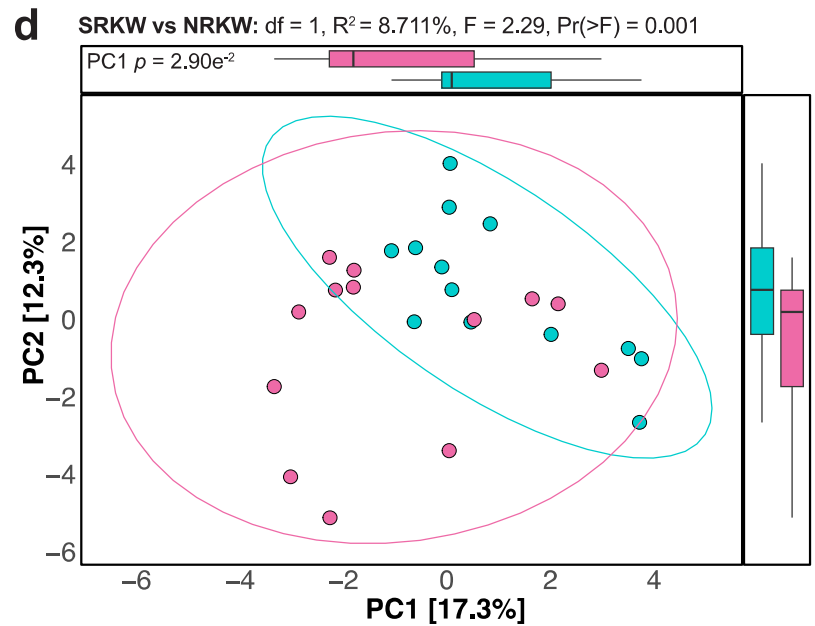
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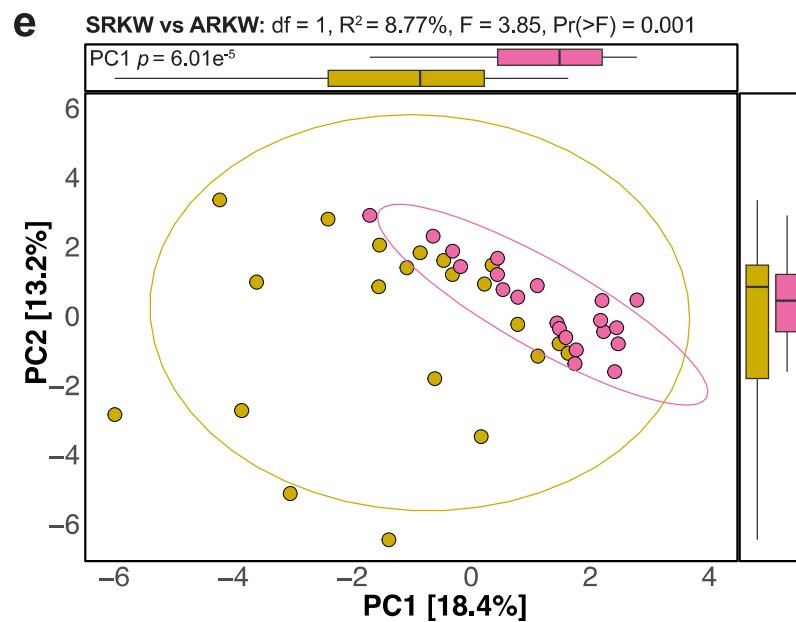
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d



e



f

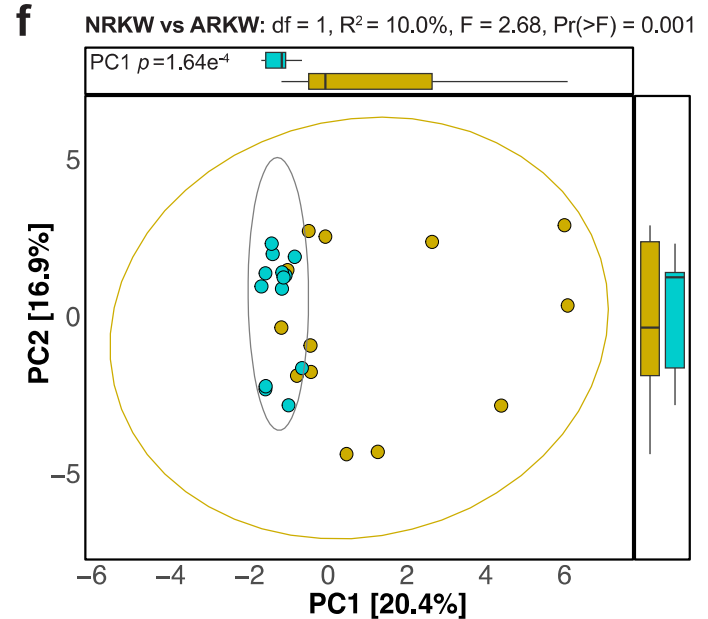
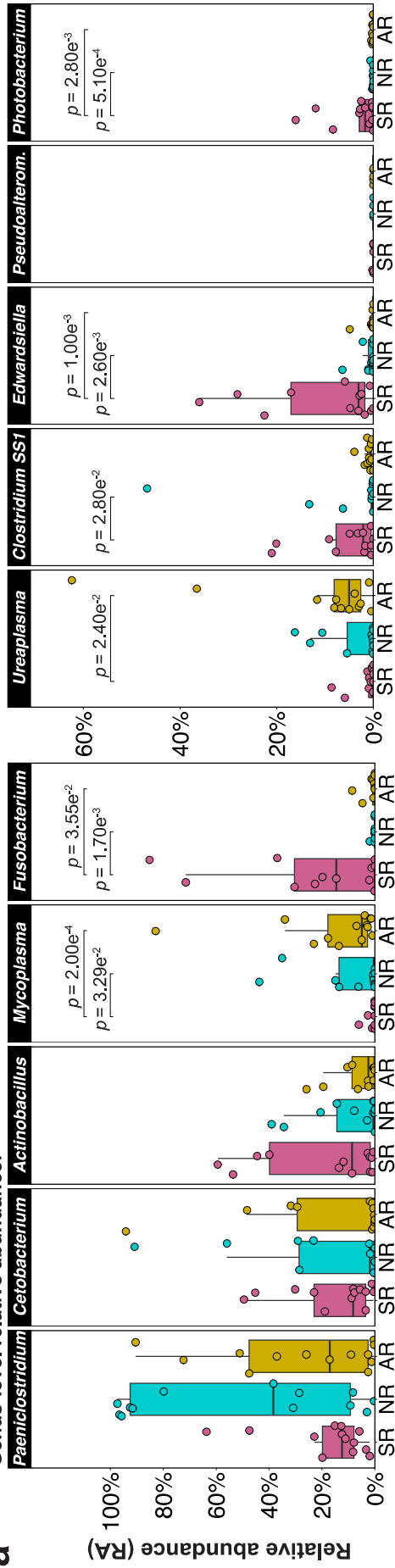


Figure 1

a Genus-level relative abundance:



b Differentially abundant ASVs: Each resident population vs both others

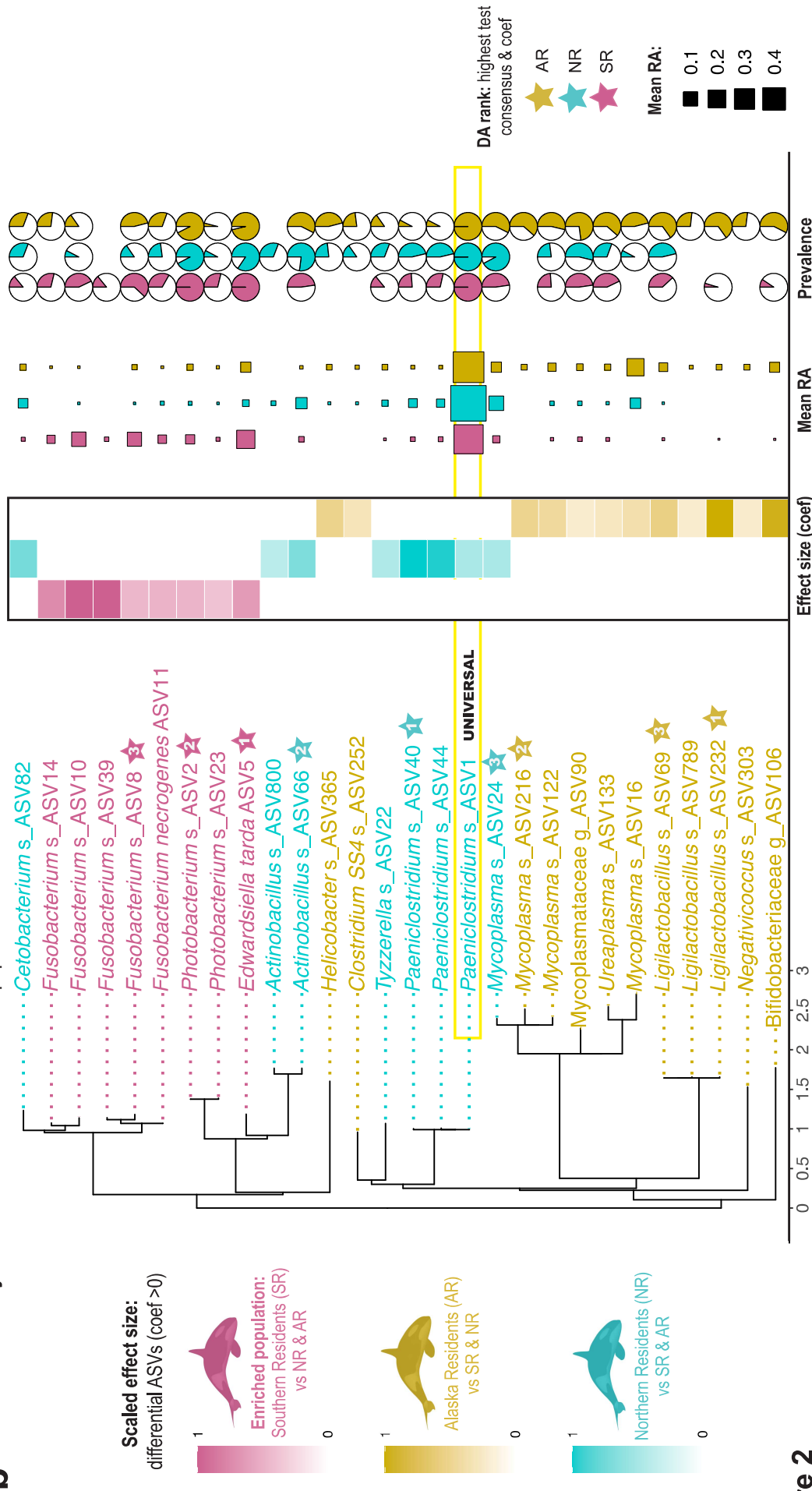
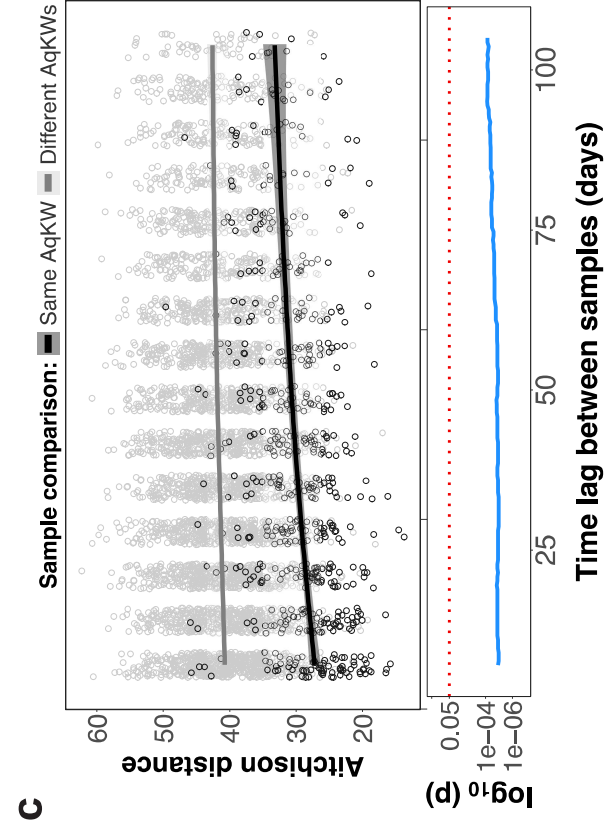
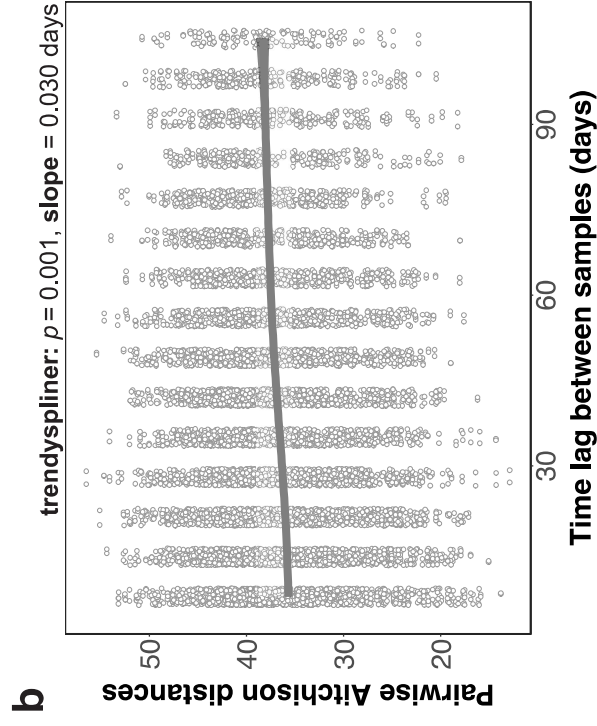
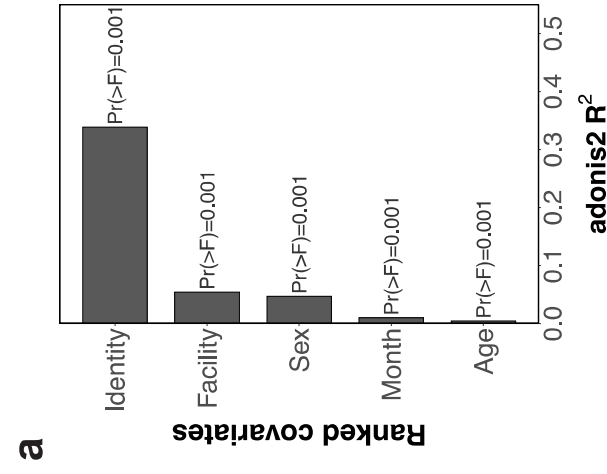


Figure 2

Aquarium-housed killer whales:



Wild Southern Resident killer whales:

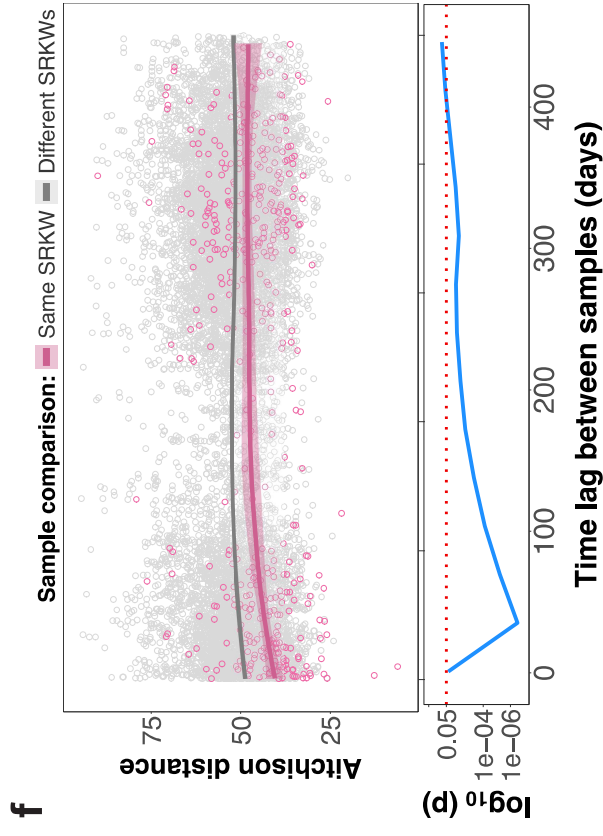
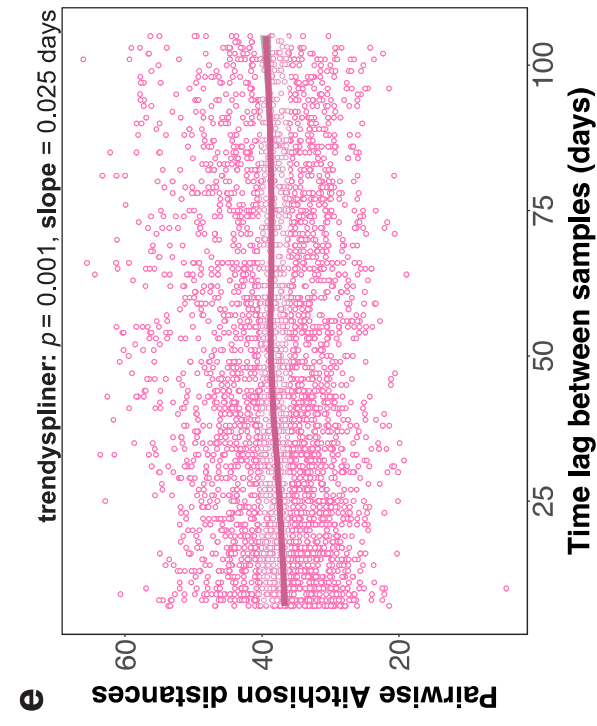
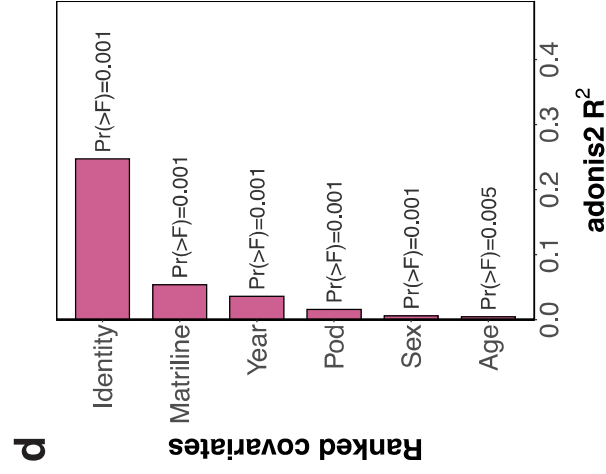


Figure 3

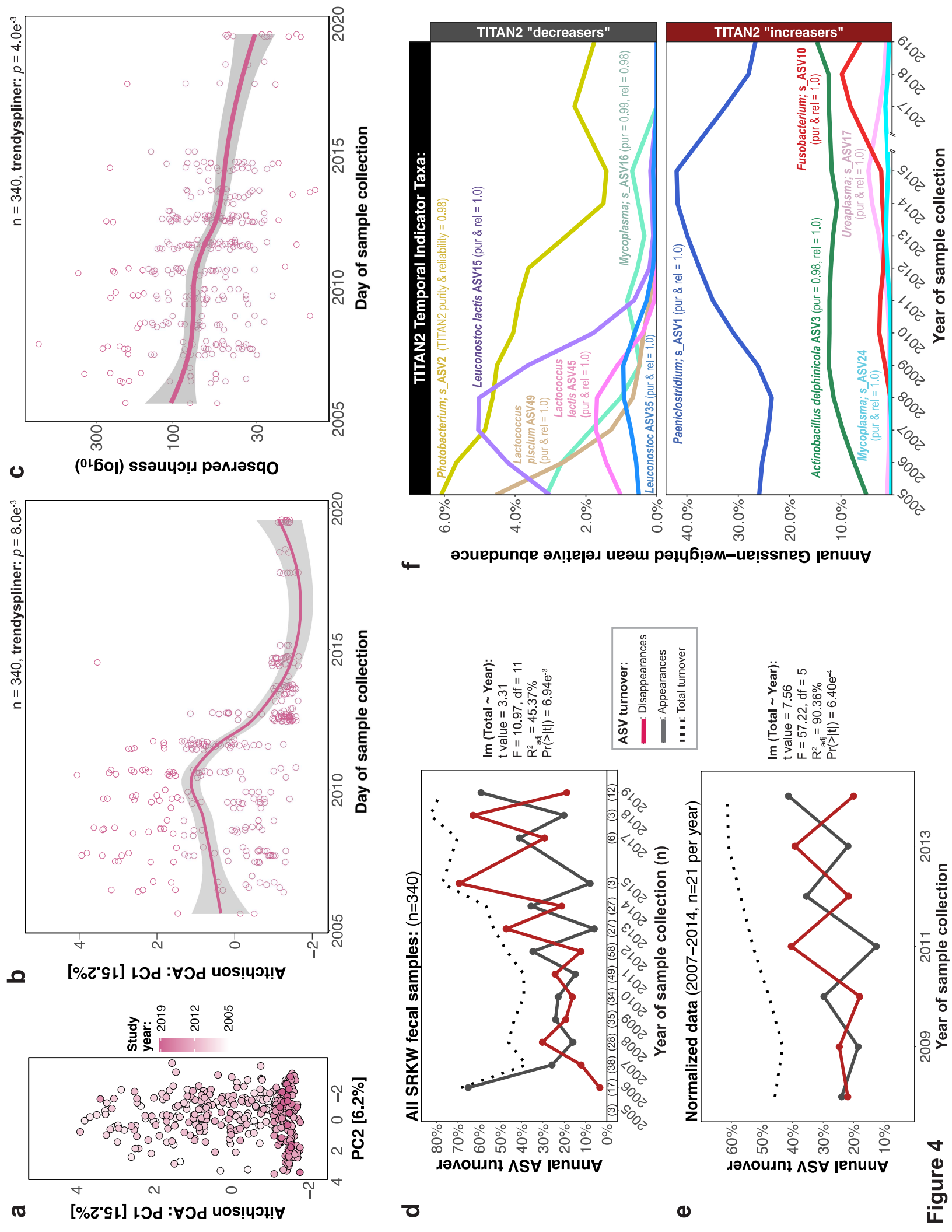


Figure 4

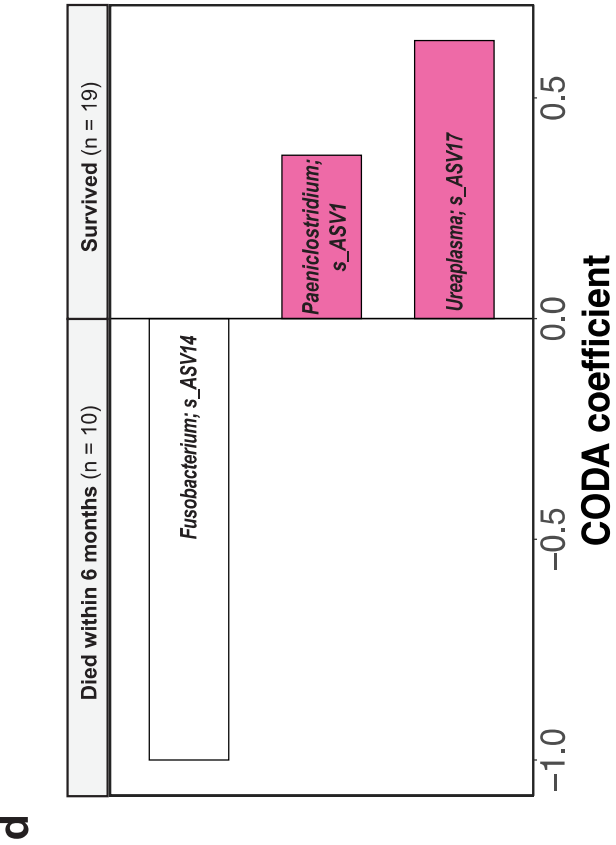
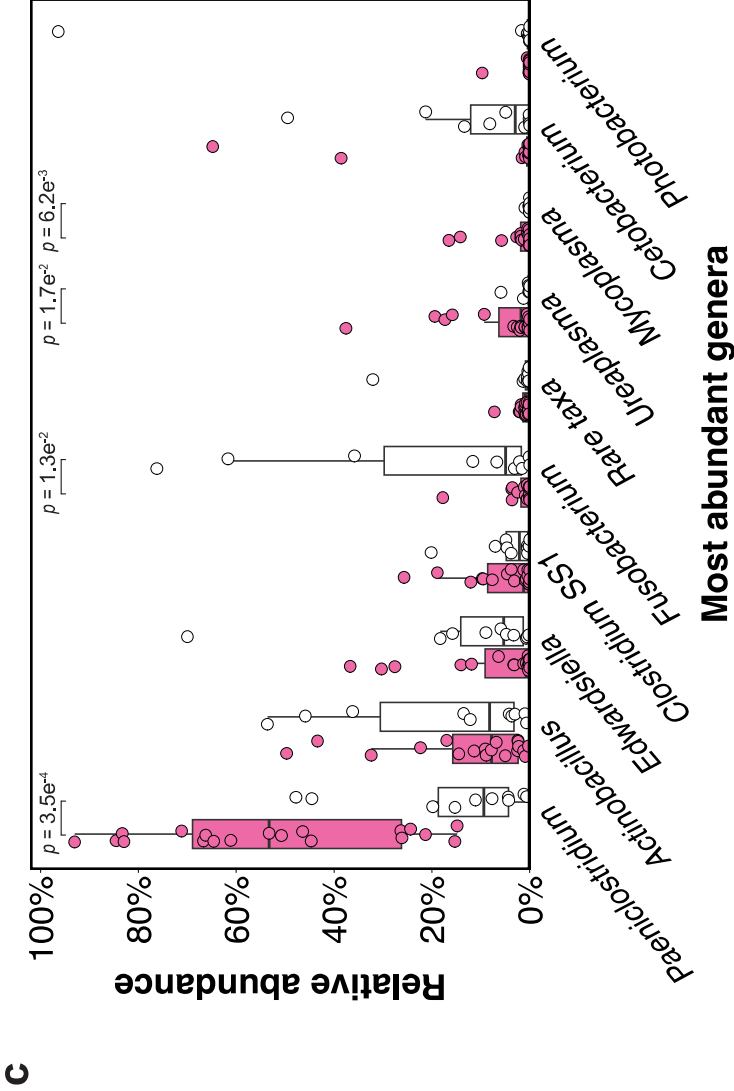
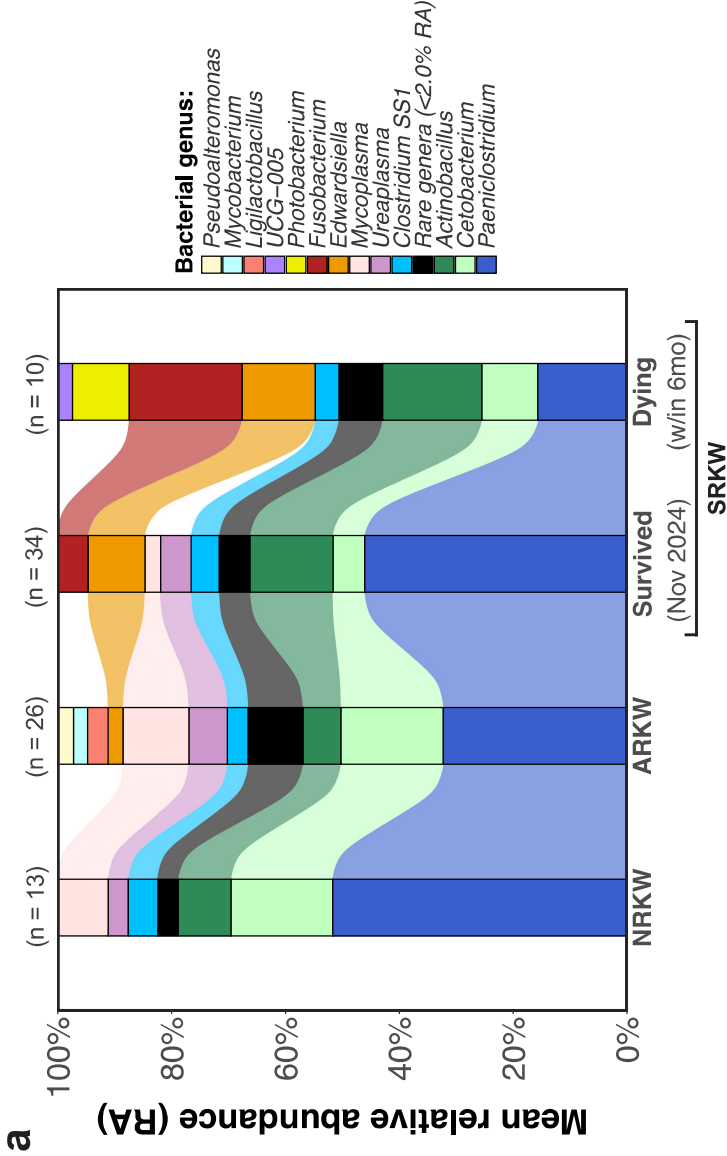


Figure 5

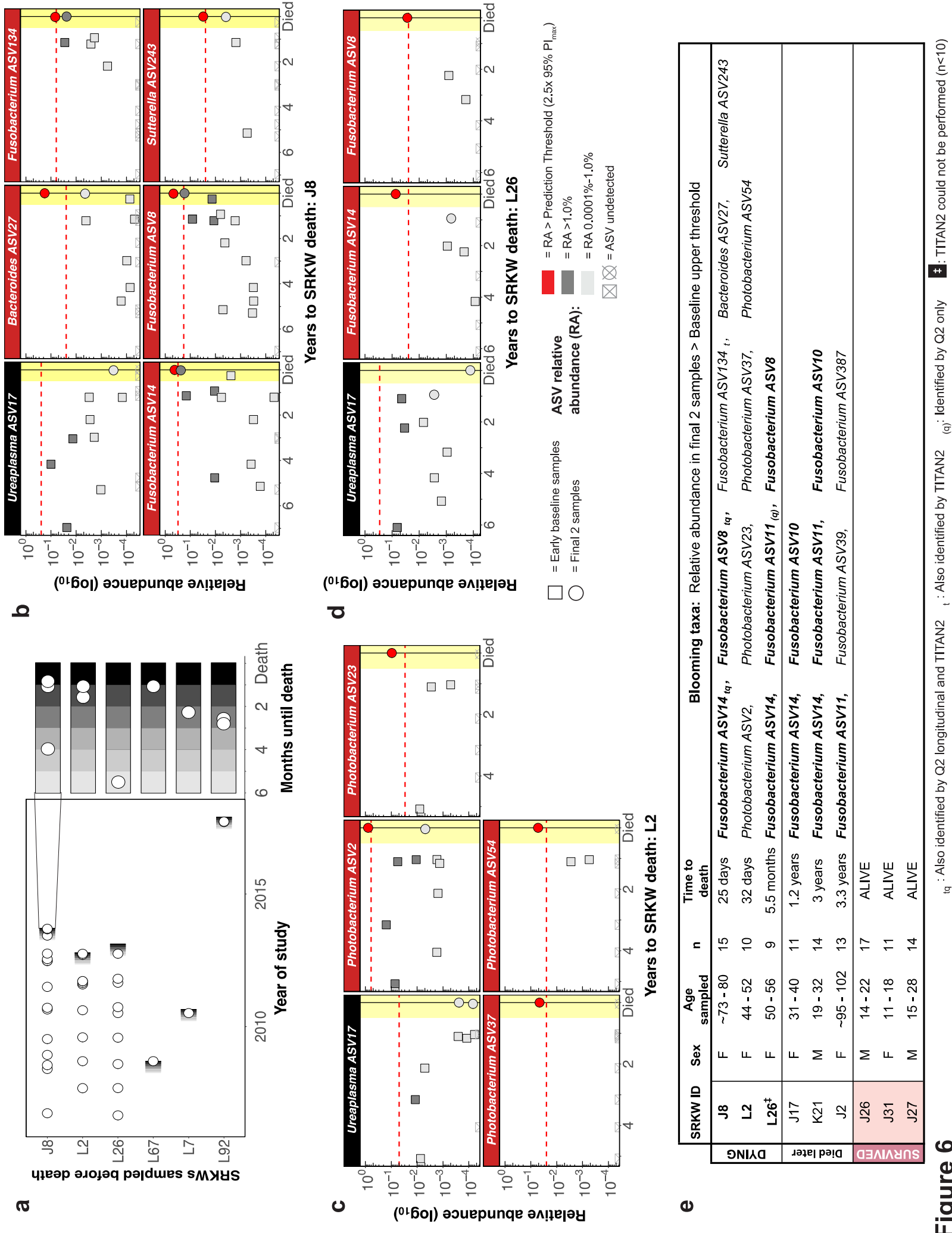


Figure 6